FORMAL ETALENESS OF A MORPHISM OF SCHEMES IS DETECTED AFFINE LOCALLY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1672-1688 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1690-1703, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally etale morphisms of schemes, formally etale ring maps, and affine open covers; the local-to-global direction is retained.

Source reference. Original lemma lemma-affine-formally-etale: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1655–1661. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-characterize-formally-etale: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1595–1604. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-formally-etale-on-opens: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1574–1578. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universal-thickening-localize: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1310–1319. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-differentials-restrict-open: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1703–1703. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. The following are equivalent:*

(1) *f is formally étale,*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(2) *for every $x \in X$ there exist opens $x \in U \subset X$ and $f(x) \in V \subset Y$ with $f(U) \subset V$ such that $f|_U : U \to V$ is formally étale,*
(3) *for every pair of affine opens $U \subset X$ and $V \subset Y$ with $f(U) \subset V$ the ring map $\mathcal{O}_Y(V) \to \mathcal{O}_X(U)$ is formally étale, and*
(4) *there exists an affine open covering $Y = \bigcup V_j$ and for each j an affine open covering $f^{-1}(V_j) = \bigcup U_{ji}$ such that $\mathcal{O}_Y(V) \to \mathcal{O}_X(U)$ is a formally étale ring map for all j and i.*

Proof. Combining Lemmas lemma-formally-etale-on-opens and lemma-affine-formally-etale we see that (1) $\Rightarrow$ (3). It is immediate that (3) $\Rightarrow$ (4). We have (4) $\Rightarrow$ (2) by Lemma lemma-affine-formally-etale. If (2) holds, then $\Omega_{X/S}$ and $\mathcal{C}_{X/S}$ vanish in a neighbourhood of every point by Lemma lemma-characterize-formally-etale (this also uses Morphisms, Lemma morphisms-lemma-differentials-restrict-open and Lemma lemma-universal-thickening-localize) whereupon Lemma lemma-characterize-formally-etale tells us that f is formally unramified. $\square$